

# Escuela Politécnica Nacional

**Nombre:** Kevin Eduardo Garcia Rodriguez

**Tema:** [Taller 3b] Gauss Jacobi y Gauss Seidel

**Repositorio GIT:** <https://github.com/Nattyrd/Metodos-Numericos-2025B>

```
In [5]: %load_ext autoreload
%autoreload 2
from src import eliminacion_gaussiana, gauss_jacobi, gauss_seidel
import numpy as np
```

```
[01-20 20:59:39][INFO][NattyrdGT] 2026-01-20 20:59:39.066448
[01-20 20:59:39][INFO][NattyrdGT] 2026-01-20 20:59:39.069920
[01-20 20:59:39][INFO][NattyrdGT] 2026-01-20 20:59:39.069920
```

## EJERCICIO 1

 Descripción

### 1.1 Metodo Gauss Seidel

```
In [7]: x=[0,0]
A=np.array([[1, 1],
            [-2, 5]], dtype=float)
b=np.array([7, 0], dtype=float)
n=2
solution, historial_x_solutions = gauss_seidel(A=A, b=b, x=[0] * n, tol=1e-5, ma
```

```
[01-20 21:00:02][INFO][NattyrdGT] i= 0 x: [[0. 0.]]
[01-20 21:00:02][INFO][NattyrdGT] i= 1 x: [[7. 2.8]]
[01-20 21:00:02][INFO][NattyrdGT] i= 2 x: [[4.2 1.68]]
[01-20 21:00:02][INFO][NattyrdGT] i= 3 x: [[5.32 2.128]]
[01-20 21:00:02][INFO][NattyrdGT] i= 3 x: [[5.32 2.128]]
[01-20 21:00:02][INFO][NattyrdGT] i= 4 x: [[4.872 1.9488]]
[01-20 21:00:02][INFO][NattyrdGT] i= 5 x: [[5.0512 2.02048]]
[01-20 21:00:02][INFO][NattyrdGT] i= 6 x: [[4.97952 1.991808]]
[01-20 21:00:02][INFO][NattyrdGT] i= 7 x: [[5.008192 2.0032768]]
[01-20 21:00:02][INFO][NattyrdGT] i= 8 x: [[4.9967232 1.99868928]]
[01-20 21:00:02][INFO][NattyrdGT] i= 9 x: [[5.00131072 2.00052429]]
[01-20 21:00:02][INFO][NattyrdGT] i= 10 x: [[4.99947571 1.99979028]]
```

```
In [36]: n2=1
solution_seidel, historial_x_solutions_Seidel_S2 = gauss_seidel(A=A, b=b, x=[100
```

```
[01-20 17:43:15][INFO][NattyrdGT] i= 0 x: [[100.  20.]]
[01-20 17:43:15][INFO][NattyrdGT] i= 1 x: [[-13.   -5.2]]
[01-20 17:43:15][INFO][NattyrdGT] i= 2 x: [[12.2   4.88]]
[01-20 17:43:16][INFO][NattyrdGT] i= 3 x: [[2.12   0.848]]
[01-20 17:43:16][INFO][NattyrdGT] i= 4 x: [[6.152  2.4608]]
[01-20 17:43:16][INFO][NattyrdGT] i= 5 x: [[4.5392  1.81568]]
[01-20 17:43:16][INFO][NattyrdGT] i= 6 x: [[5.18432  2.073728]]
[01-20 17:43:16][INFO][NattyrdGT] i= 7 x: [[4.926272  1.9705088]]
[01-20 17:43:16][INFO][NattyrdGT] i= 8 x: [[5.0294912  2.01179648]]
[01-20 17:43:16][INFO][NattyrdGT] i= 9 x: [[4.98820352  1.99528141]]
[01-20 17:43:16][INFO][NattyrdGT] i= 10 x: [[5.00471859  2.00188744]]
```

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

```
In [29]: import numpy as np
import matplotlib.pyplot as plt

def graficar_historial_x_nuevo(historial_x):
    """
    Grafica la trayectoria de las soluciones en el plano X1 vs X2
    mostrando cómo se mueven los valores en cada iteración.

    Parameters
    -----
    historial_x : list
        Lista de vectores x obtenidos en cada iteración.
        Cada x debe ser de la forma [x1, x2]
    """

    historial_x = np.array(historial_x)

    if historial_x.shape[1] != 2:
        raise ValueError("Este método solo funciona para vectores de 2 variables")

    x1 = historial_x[:, 0]
    x2 = historial_x[:, 1]

    plt.plot(x1, x2, marker='o')

    # Etiquetar cada punto con la iteración
    for i in range(len(x1)):
        plt.text(x1[i], x2[i], f'{i}', fontsize=9)

    plt.xlabel('X1')
    plt.ylabel('X2')
    plt.title('Trayectoria del vector solución (X1 vs X2)')
    plt.grid(True)
    plt.show()
```

```
In [58]: def graficar_historial_x(historial_x):
    """
    Grafica la evolución de cada componente del vector x (de soluciones)
    a lo largo de las iteraciones.

    Parameters
    -----
    historial_x : list
        Lista de vectores x obtenidos en cada iteración
```

```

historial_x = np.array(historial_x) # shape: (iteraciones, n)

iteraciones = range(len(historial_x))
n_variables = historial_x.shape[1]

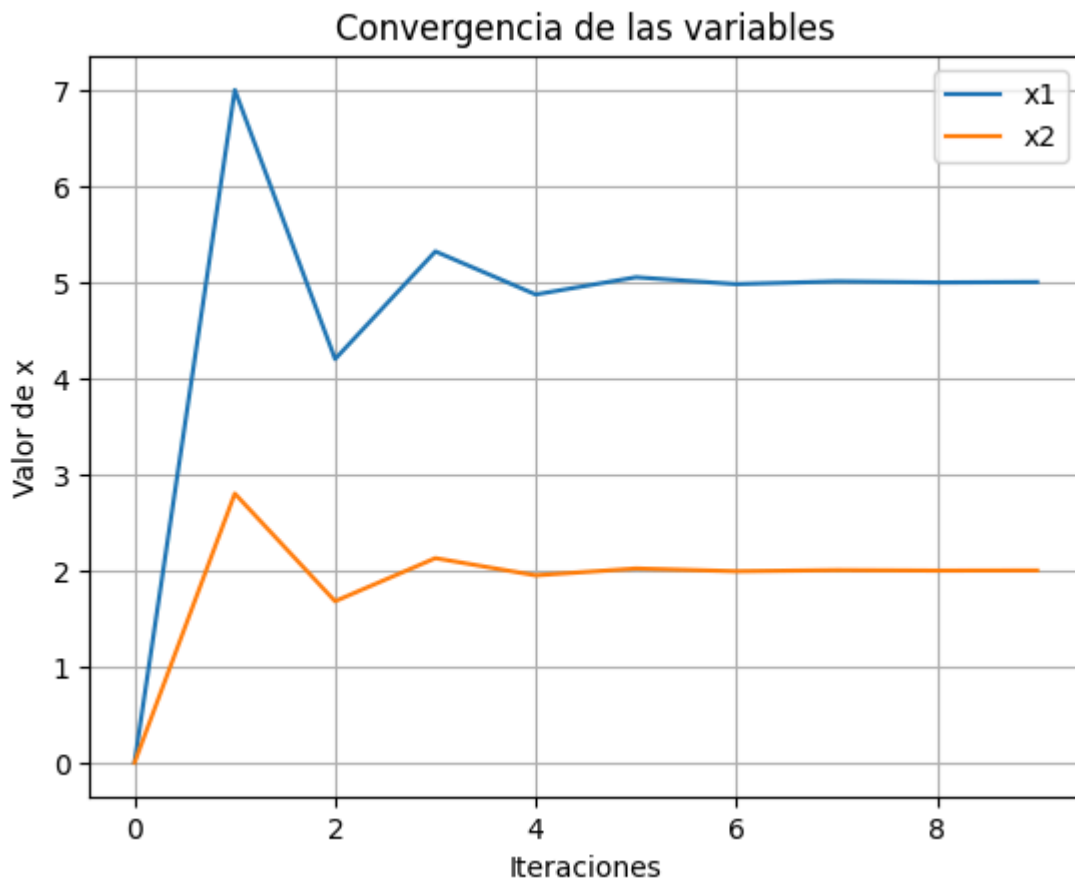
for i in range(n_variables):
    plt.plot(iteraciones, historial_x[:, i], label=f'x{i+1}')

plt.xlabel('Iteraciones')
plt.ylabel('Valor de x')
plt.title('Convergencia de las variables ')
plt.legend()
plt.grid(True)
plt.show()

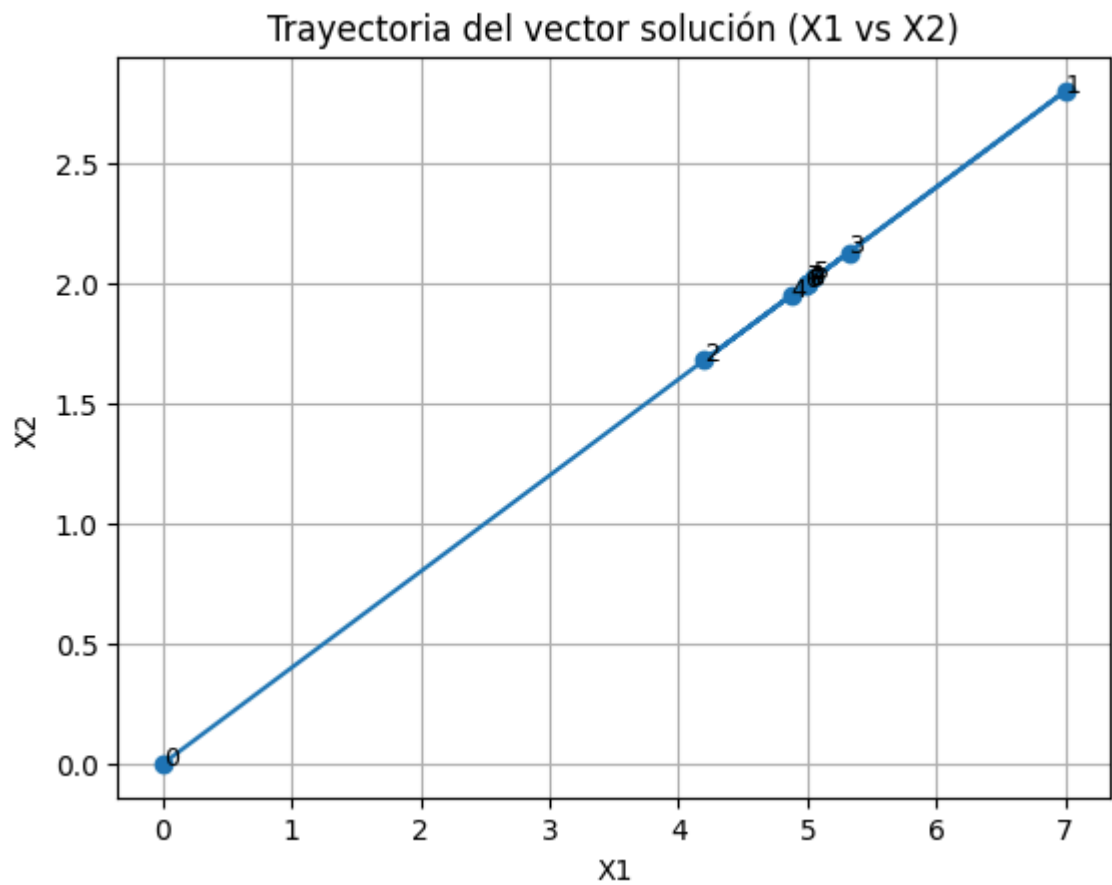
```

## Graficar Con Gauss Seidel

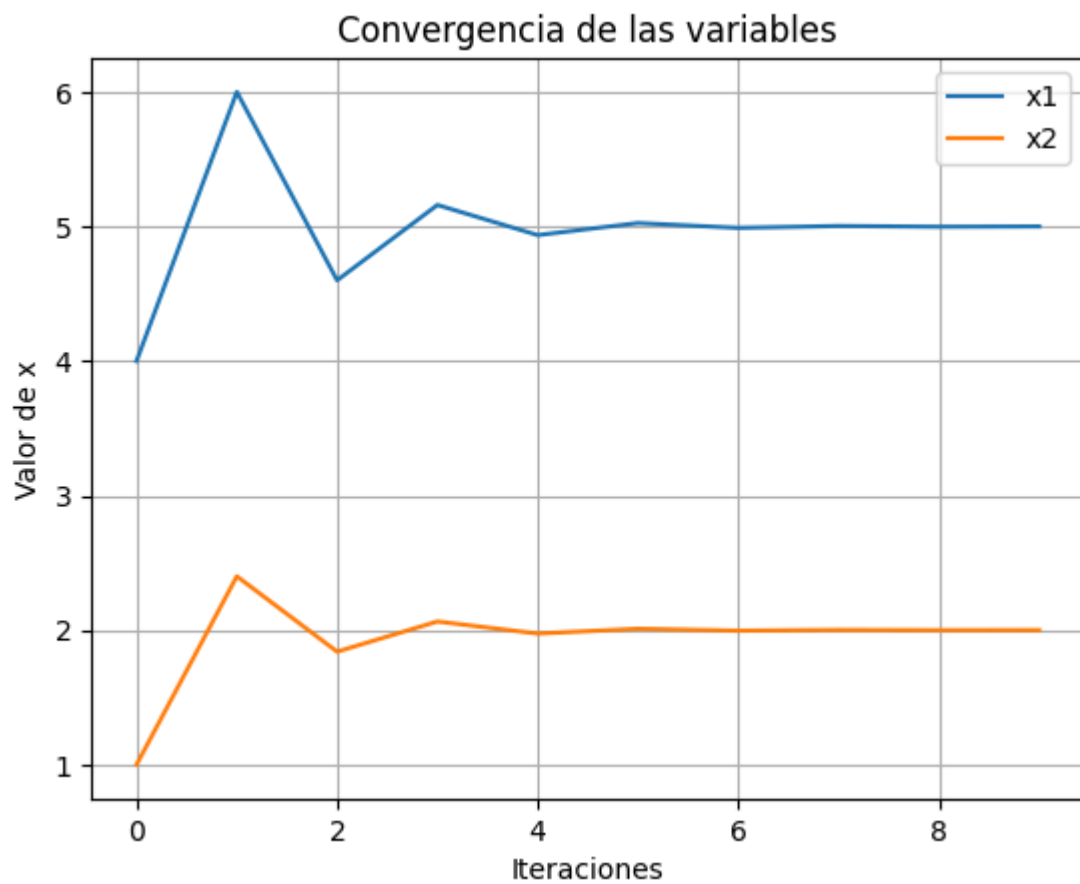
In [39]: `graficar_historial_x(historial_x_solutions)`



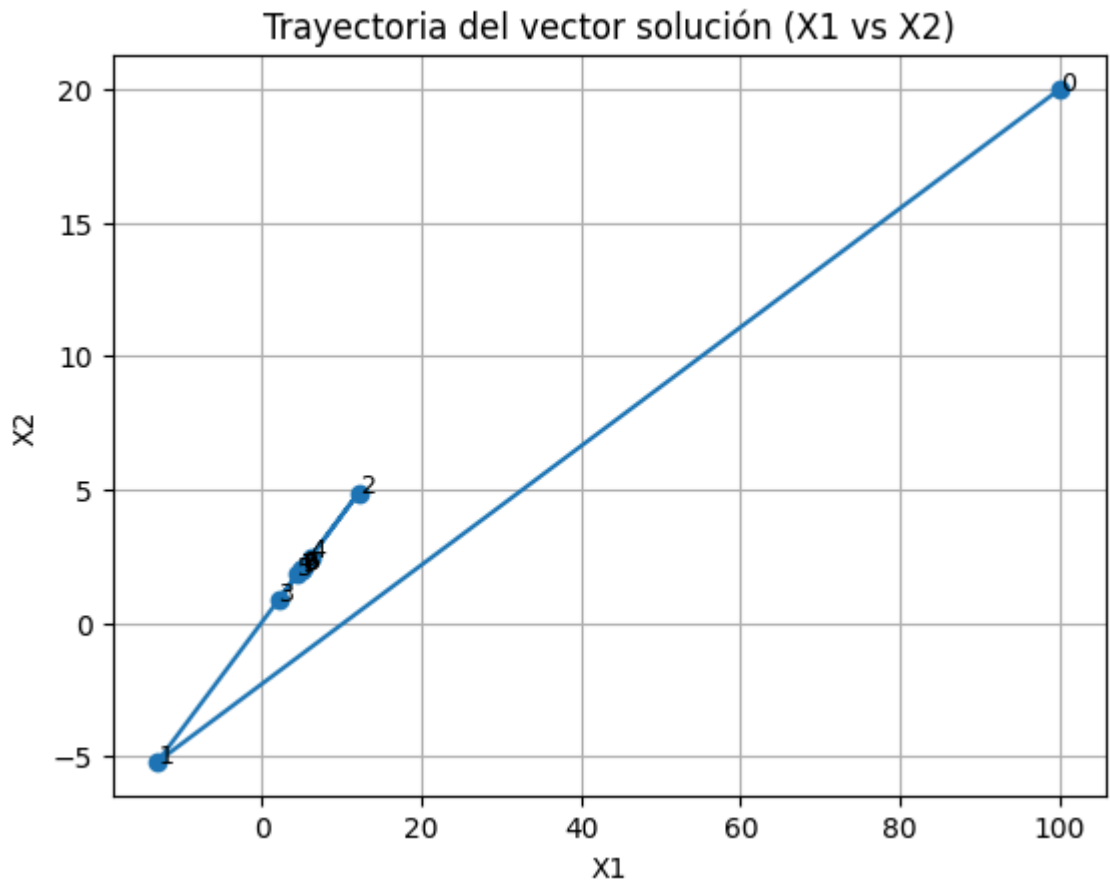
In [42]: `graficar_historial_x_nuevo(historial_x_solutions)`



In [23]: `graficar_historial_x(historial_x_solutions_Seidel_S2)`



In [43]: `graficar_historial_x_nuevo(historial_x_solutions_Seidel_S2)`

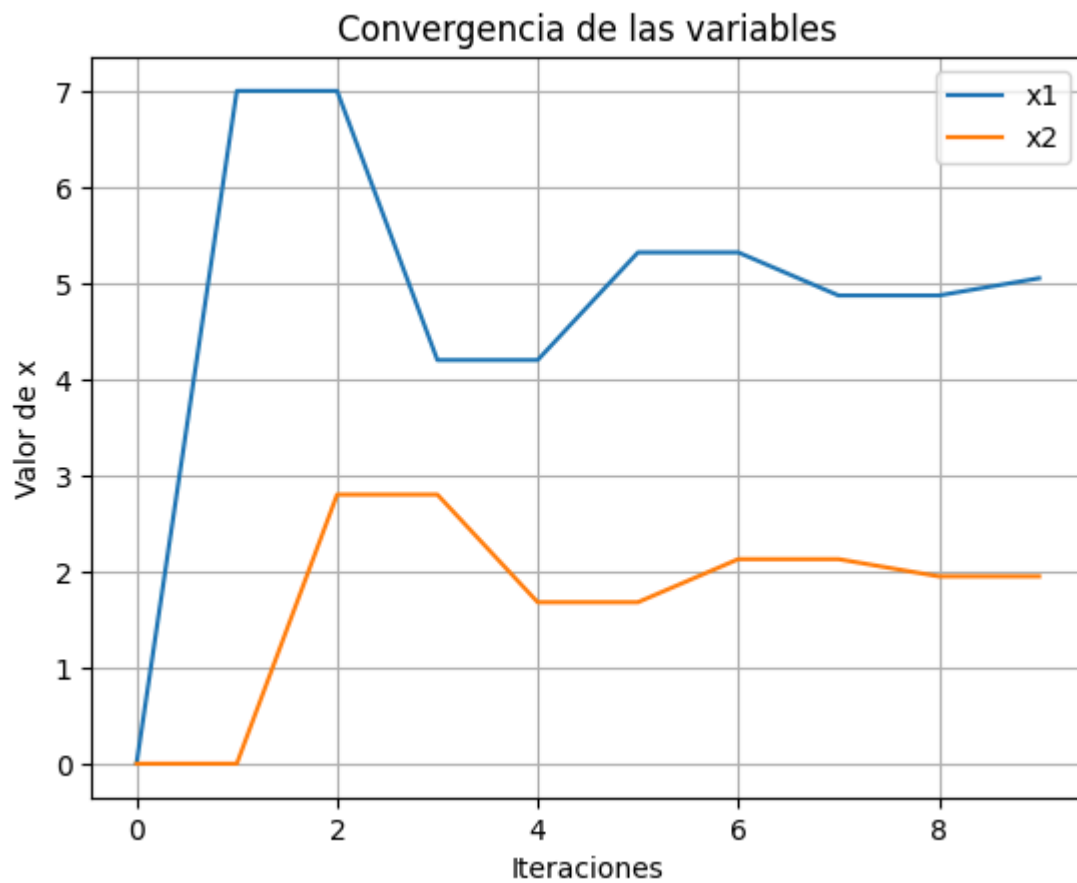


## 1.2 Metodo de Gauss Jacobi

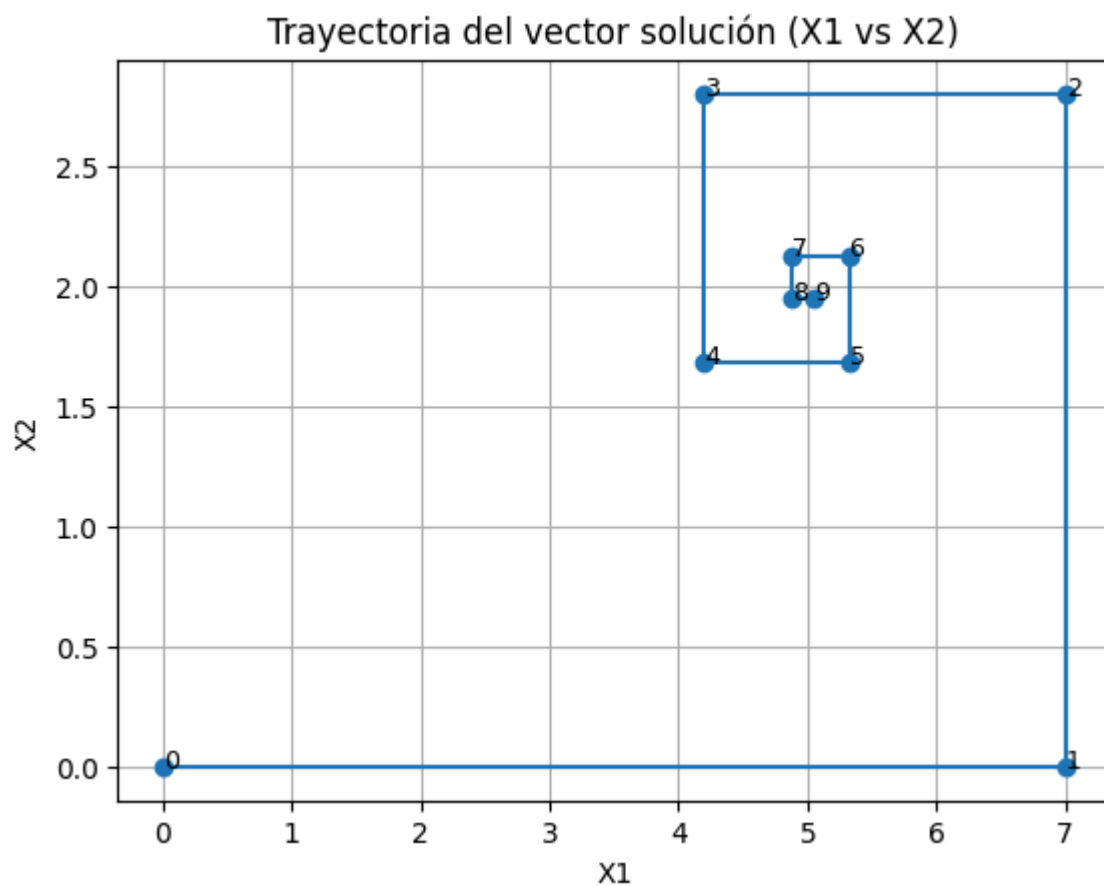
```
In [10]: solution_jacobi, historial_x_solutions_jacobi = gauss_jacobi(A=A, b=b, x=[0] * n)

[01-20 21:00:22][INFO][NattyrdGT] i= 0 x: [[0. 0.]]
[01-20 21:00:22][INFO][NattyrdGT] i= 10 x: [[5.0512  2.02048]]
[01-20 21:00:22][INFO][NattyrdGT] i= 10 x: [[5.0512  2.02048]]
```

```
In [25]: graficar_historial_x(historial_x_solutions_jacobi)
```



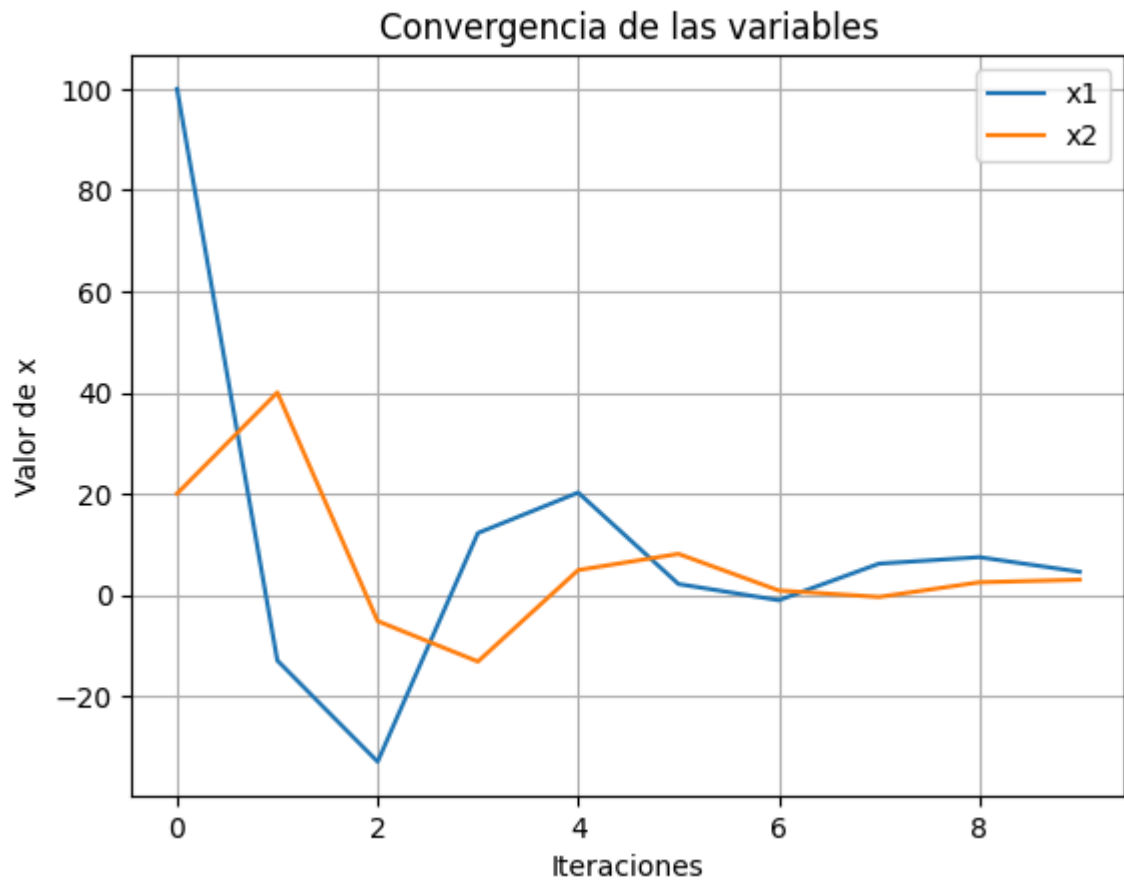
In [44]: `graficar_historial_x_nuevo(historial_x_solutions_jacobi)`



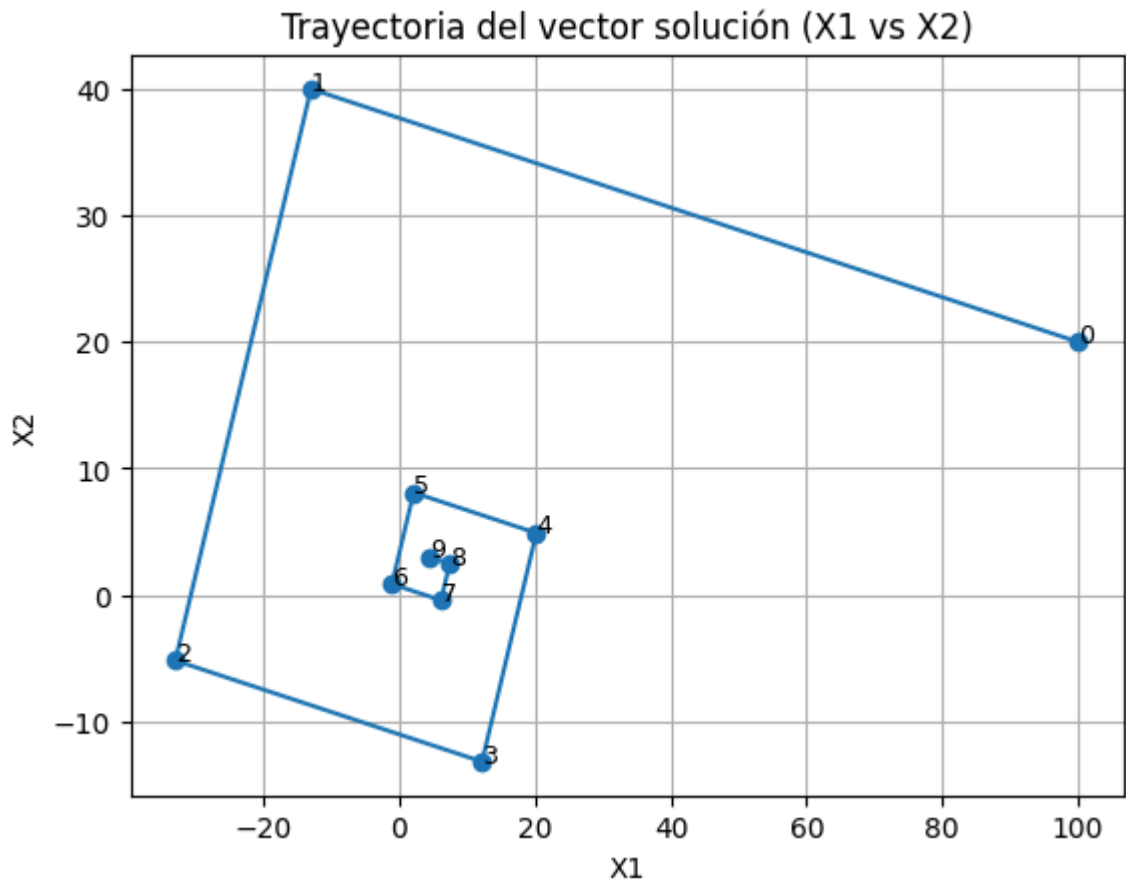
In [14]: `n1=1  
solution_jacobi_S2, historial_x_solutions_jacobi_S2 = gauss_jacobi(A=A, b=b, x=`

```
[01-20 21:01:44][INFO][NattyrdGT] i= 0 x: [[4. 2.]]  
[01-20 21:01:44][INFO][NattyrdGT] i= 10 x: [[5.01024 2.  ]]  
[01-20 21:01:44][INFO][NattyrdGT] i= 10 x: [[5.01024 2.  ]]
```

```
In [32]: graficar_historial_x(historial_x_solutions_jacobi_S2)
```



```
In [33]: graficar_historial_x_nuevo(historial_x_solutions_jacobi_S2)
```



Como es la respuesta, Entonces no tenemos puntos que graficar ya que lo encuentra en la primera iteracion.

```
In [11]: import numpy as np
import matplotlib.pyplot as plt

def graficar_multiples_historiales(historiales, etiquetas):
    """
    Grafica varias trayectorias X1 vs X2 en un mismo plano
    para comparar distintos métodos o condiciones iniciales.

    Parameters
    -----
    historiales : list
        Lista de historiales_x.
        Cada historial_x es una lista de vectores [x1, x2]

    etiquetas : list
        Lista de etiquetas (nombres) para cada historial
        Ej: ["Jacobi", "Gauss-Seidel"]
    """

    if len(historiales) != len(etiquetas):
        raise ValueError("La cantidad de historiales y etiquetas debe coincidir")

    for historial_x, label in zip(historiales, etiquetas):
        historial_x = np.array(historial_x)

        if historial_x.shape[1] != 2:
            raise ValueError("Cada historial debe ser de 2 variables (x1, x2)")
```



```

x1 = historial_x[:, 0]
x2 = historial_x[:, 1]

plt.plot(x1, x2, marker='o', label=label)

# Etiquetar cada iteración
for i in range(len(x1)):
    plt.text(x1[i], x2[i], f'{i}', fontsize=8)

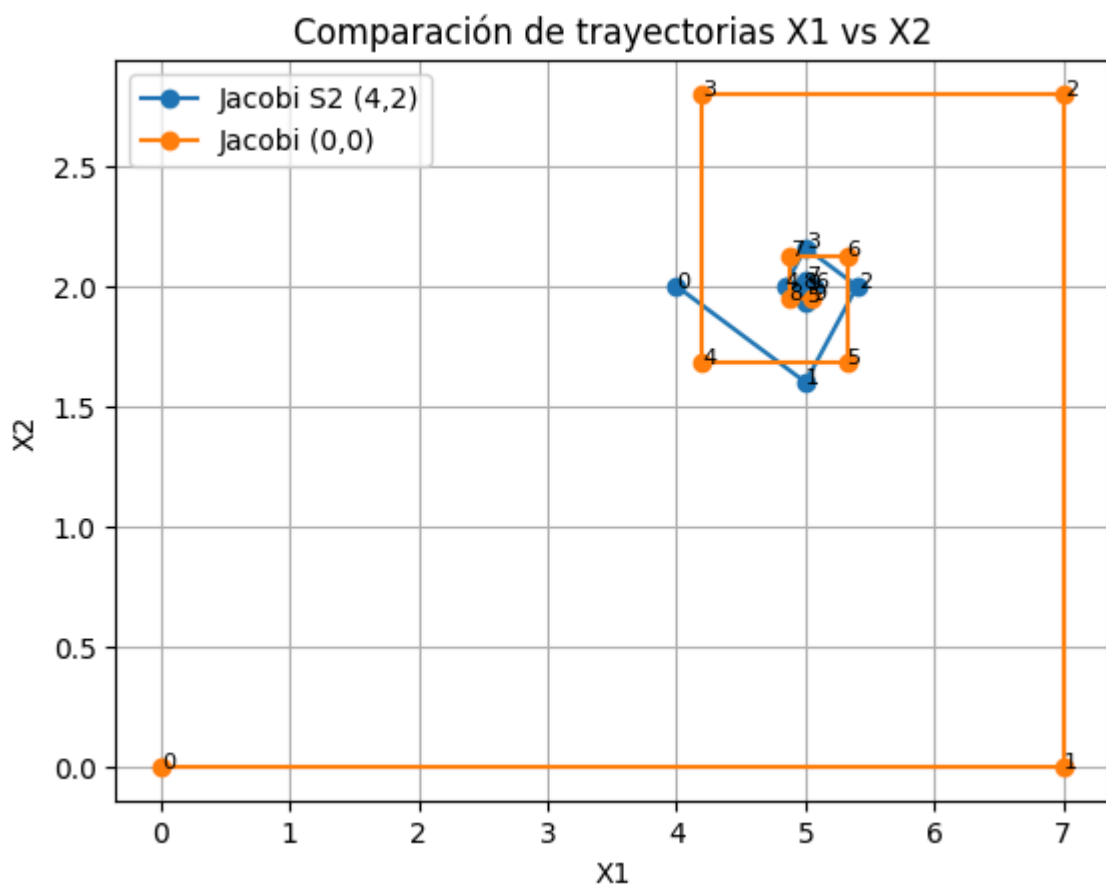
plt.xlabel('X1')
plt.ylabel('X2')
plt.title('Comparación de trayectorias X1 vs X2')
plt.legend()
plt.grid(True)
plt.show()

```

```

In [16]: graficar_multiples_historiales(
        [historial_x_solutions_jacobi_S2, historial_x_solutions_jacobi],
        ["Jacobi S2 (4,2)", "Jacobi (0,0)"]
    )

```



## EJERCICIO 2

 Descripción

## Metodo Gauss-Jacobi

```

In [39]: A=np.array([[1, 1],
                    [-2, 1]], dtype=float)
        b=np.array([6, 0], dtype=float)

```

```
n=1
solution22, historial_x_solutions_jacobi2 = gauss_jacobi(A=A, b=b, x=[3,4] * n,
```

```
[01-20 21:33:06][INFO][NattyrdGT] i= 0 x: [[3. 4.]]
[01-20 21:33:06][INFO][NattyrdGT] i= 10 x: [[-30.  4.]]
[01-20 21:33:06][INFO][NattyrdGT] i= 10 x: [[-30.  4.]]
```

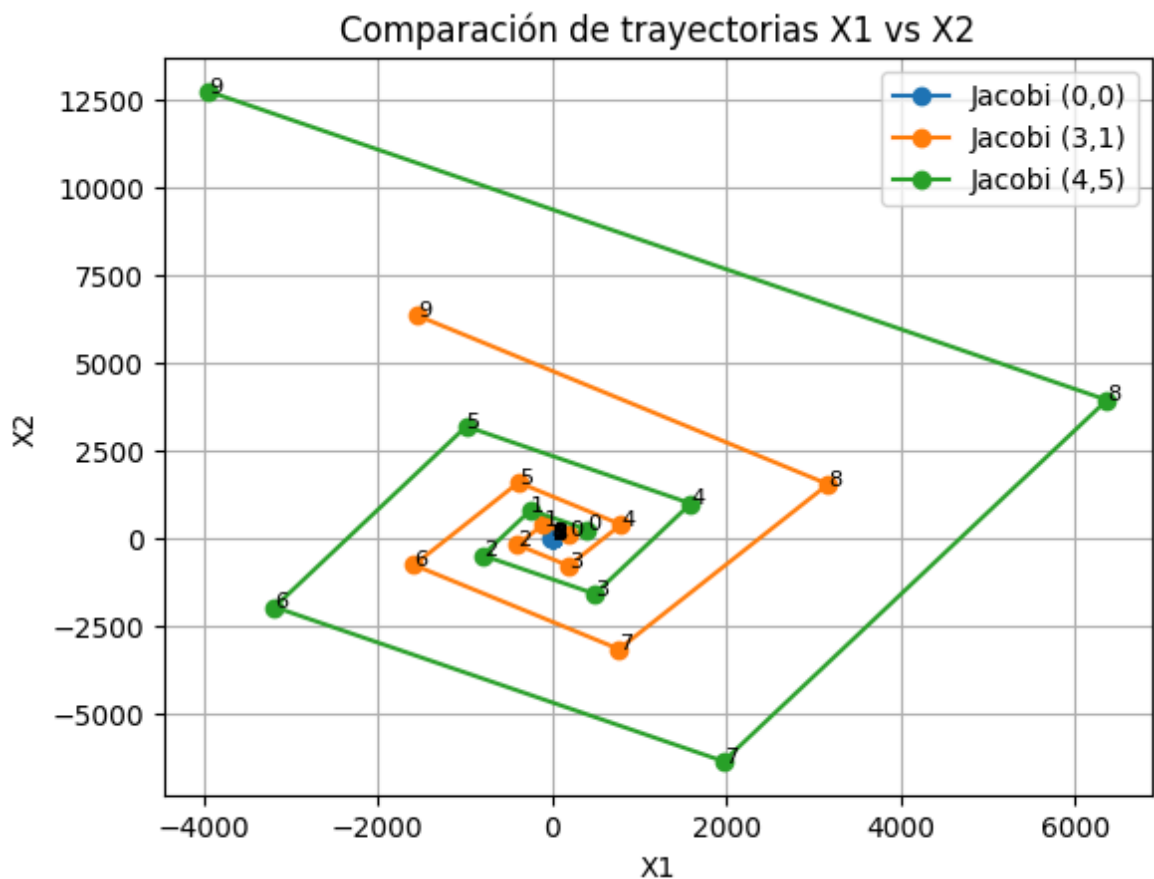
```
In [40]: A=np.array([[1, 1],
                    [-2, 1]], dtype=float)
b=np.array([6, 0], dtype=float)
n=1
solution221, historial_x_solutions_jacobi21 = gauss_jacobi(A=A, b=b, x=[200,100]
```

```
[01-20 21:33:09][INFO][NattyrdGT] i= 0 x: [[200. 100.]]
[01-20 21:33:09][INFO][NattyrdGT] i= 10 x: [[-6334. -3068.]]
[01-20 21:33:09][INFO][NattyrdGT] i= 10 x: [[-6334. -3068.]]
```

```
In [41]: A=np.array([[1, 1],
                    [-2, 1]], dtype=float)
b=np.array([6, 0], dtype=float)
n=1
solution222, historial_x_solutions_jacobi22 = gauss_jacobi(A=A, b=b, x=[400,250]
```

```
[01-20 21:33:11][INFO][NattyrdGT] i= 0 x: [[400. 250.]]
[01-20 21:33:11][INFO][NattyrdGT] i= 10 x: [[-12734. -7868.]]
[01-20 21:33:11][INFO][NattyrdGT] i= 10 x: [[-12734. -7868.]]
```

```
In [42]: graficar_multiples_historiales(
        [historial_x_solutions_jacobi2, historial_x_solutions_jacobi21, historial_x_
        ["Jacobi (0,0)", "Jacobi (3,1)", "Jacobi (4,5)"]
    )
```



## Metodo Gauss-Seidel

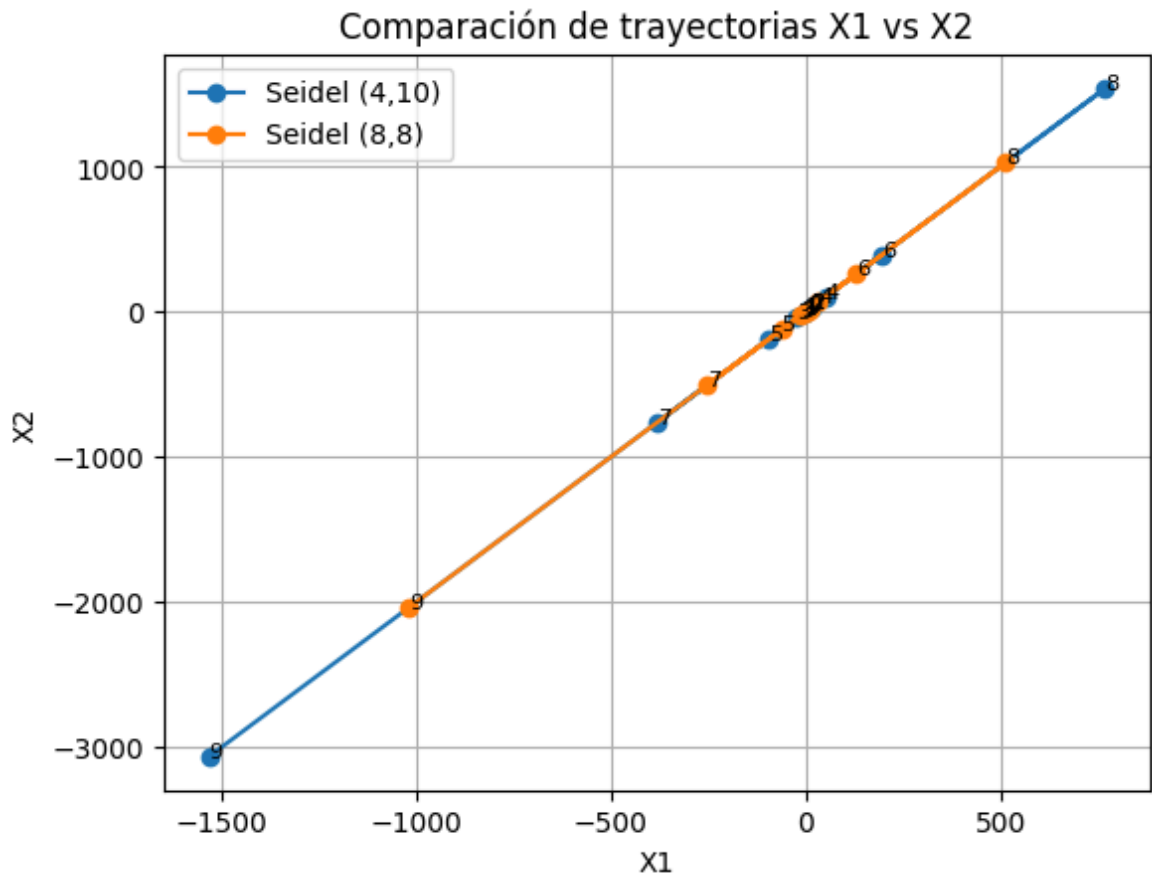
```
In [47]: A=np.array([[1, 1],
                    [-2, 1]], dtype=float)
b=np.array([6, 0], dtype=float)
n=1
solution212, historial_x_solutions_seidel21 = gauss_seidel(A=A, b=b, x=[4,10] *
```

```
[01-20 21:38:22][INFO][NattyrdGT] i= 0 x: [[ 4. 10.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 1 x: [[-4. -8.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 2 x: [[14. 28.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 3 x: [[-22. -44.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 4 x: [[ 50. 100.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 5 x: [[ -94. -188.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 6 x: [[194. 388.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 7 x: [[-382. -764.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 8 x: [[ 770. 1540.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 3 x: [[-22. -44.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 4 x: [[ 50. 100.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 5 x: [[ -94. -188.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 6 x: [[194. 388.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 7 x: [[-382. -764.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 8 x: [[ 770. 1540.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 9 x: [[-1534. -3068.]]
[01-20 21:38:22][INFO][NattyrdGT] i= 10 x: [[3074. 6148.]]
```

```
In [48]: A=np.array([[1, 1],
                    [-2, 1]], dtype=float)
b=np.array([6, 0], dtype=float)
n=1
solution212, historial_x_solutions_seidel211 = gauss_seidel(A=A, b=b, x=[8,8] *
```

```
[01-20 21:38:24][INFO][NattyrdGT] i= 0 x: [[8. 8.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 1 x: [[-2. -4.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 2 x: [[10. 20.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 3 x: [[-14. -28.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 4 x: [[34. 68.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 5 x: [[ -62. -124.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 6 x: [[130. 260.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 7 x: [[-254. -508.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 8 x: [[ 514. 1028.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 9 x: [[-1022. -2044.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 1 x: [[-2. -4.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 2 x: [[10. 20.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 3 x: [[-14. -28.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 4 x: [[34. 68.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 5 x: [[ -62. -124.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 6 x: [[130. 260.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 7 x: [[-254. -508.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 8 x: [[ 514. 1028.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 9 x: [[-1022. -2044.]]
[01-20 21:38:24][INFO][NattyrdGT] i= 10 x: [[2050. 4100.]]
```

```
In [49]: graficar_múltiples_historiales(
        [historial_x_solutions_seidel21, historial_x_solutions_seidel211],
        ["Seidel (4,10)", "Seidel (8,8)"]
    )
```



## EJERCICIO 3

 Descripción

## Metodo de Gauss-Seidel

```
In [73]: import numpy as np
import matplotlib.pyplot as plt

def graficar_convergencia_4x4_numerada(historial_x):
    """
    Grafica la convergencia de un sistema 4x4.
    Cada punto está numerado según la iteración.

    Parameters
    -----
    historial_x : list
        Lista de vectores x (cada uno de tamaño 4)
    """

    historial_x = np.array(historial_x)

    if historial_x.shape[1] != 4:
        raise ValueError("Este método está diseñado solo para sistemas 4x4")

    iteraciones = range(len(historial_x))

    fig, axs = plt.subplots(2, 2, figsize=(11, 9))
    fig.suptitle("Convergencia del sistema 4x4 (iteraciones numeradas)")
```

```

etiquetas = ["X1", "X2", "X3", "X4"]

for i, ax in enumerate(axes.flat):
    ax.plot(iteraciones, historial_x[:, i], marker='o')

    # Numerar cada punto
    for k in iteraciones:
        ax.text(
            k,
            historial_x[k, i],
            f'{k}',
            fontsize=9,
            ha='right',
            va='bottom'
        )

    ax.set_xlabel("Iteraciones")
    ax.set_ylabel(etiquetas[i])
    ax.grid(True)

plt.tight_layout()
plt.show()

```

In [52]:

```

import numpy as np

A = np.array([
    [ 1, -1,  2,  0],
    [-1,  1, -1,  3],
    [ 2, -1,  1, -1],
    [ 0,  3, -1,  2]
], dtype=float)

b = np.array([
    1360,
    1130,
    1350,
    3650
], dtype=float)
n=1
solution2122, historial_x_solutions_seidel2112 = gauss_seidel(A=A, b=b, x=[4,2,3

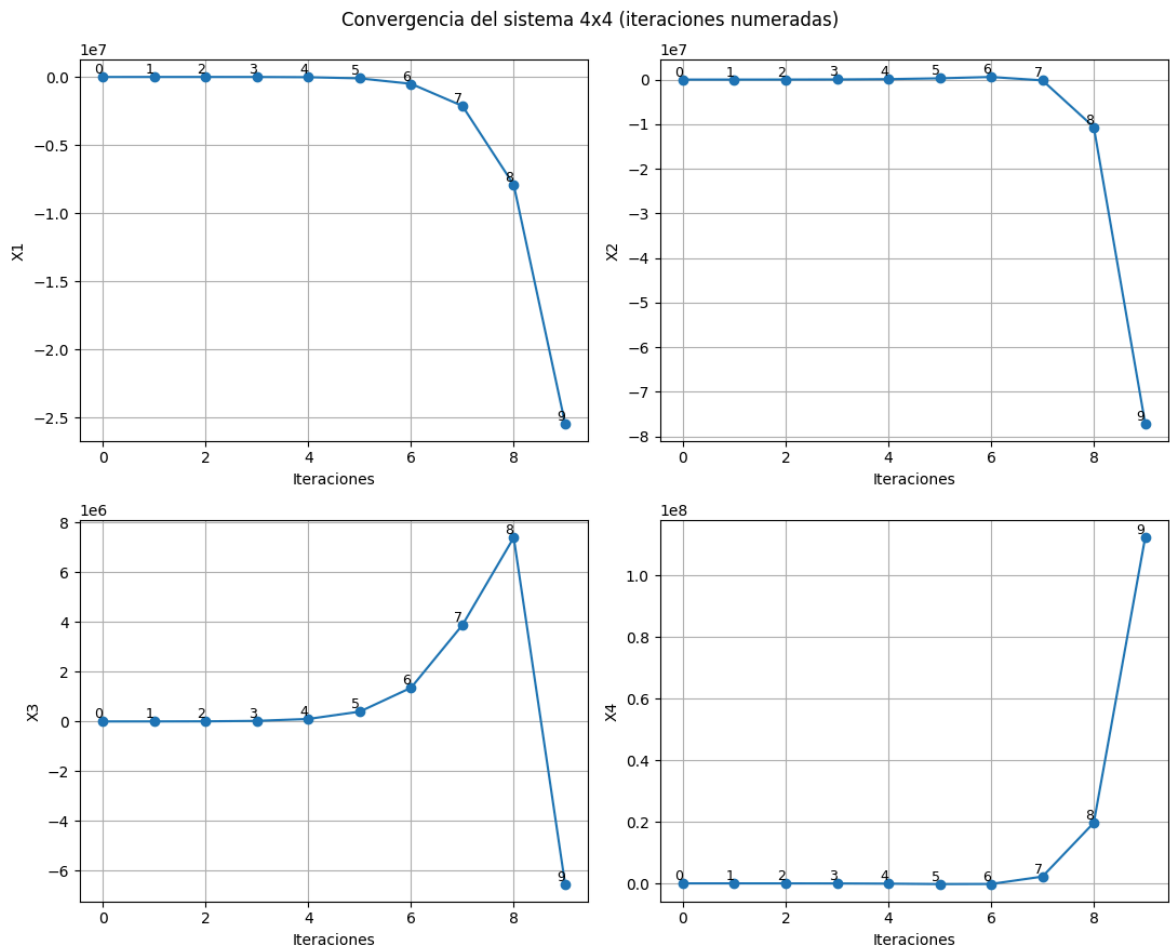
```

```

[01-20 21:43:27][INFO][NattyrdGT] i= 0 x: [[4. 2. 3. 1.]]
[01-20 21:43:27][INFO][NattyrdGT] i= 1 x: [[ 1356.   2486.   1125.  -1341.5]]
[01-20 21:43:27][INFO][NattyrdGT] i= 2 x: [[ 1596.   7875.5  4692.   -7642.25]]
[01-20 21:43:27][INFO][NattyrdGT] i= 3 x: [[ -148.5   28600.25  22605.   -297
72.875]]
[01-20 21:43:27][INFO][NattyrdGT] i= 4 x: [[-15249.75   97803.875   99880.5   -
94940.5625]]
[01-20 21:43:27][INFO][NattyrdGT] i= 5 x: [[-100597.125   285235.0625   392838.7
5   -229608.21875]]
[01-20 21:43:27][INFO][NattyrdGT] i= 6 x: [[-499082.4375   583710.96875   135361
7.625   -196932.640625]]
[01-20 21:43:27][INFO][NattyrdGT] i= 7 x: [[-2122164.28125   -176618.734375   38
72127.1875   2202816.6953125]]
[01-20 21:43:27][INFO][NattyrdGT] i= 8 x: [[ -7919513.109375   -10654706.0078125
7388486.90625
19678127.46484375]]
[01-20 21:43:27][INFO][NattyrdGT] i= 9 x: [[-2.54303198e+07 -7.70750853e+07 -6.53
496820e+06  1.12346969e+08]]
[01-20 21:43:27][INFO][NattyrdGT] i= 10 x: [[-6.40037889e+07 -4.07578534e+08 -1.6
7222637e+08  5.27758307e+08]]

```

In [79]: `graficar_convergencia_4x4_numerada(historial_x_solutions_seidel2112)`



```

In [53]: A = np.array([
    [ 1, -1, 2, 0],
    [-1, 1, -1, 3],
    [ 2, -1, 1, -1],
    [ 0, 3, -1, 2]
], dtype=float)

b = np.array([

```

```

1360,
1130,
1350,
3650
], dtype=float)
n=1
solution21221, historial_x_solutions_seidel21121 = gauss_seidel(A=A, b=b, x=[2,1

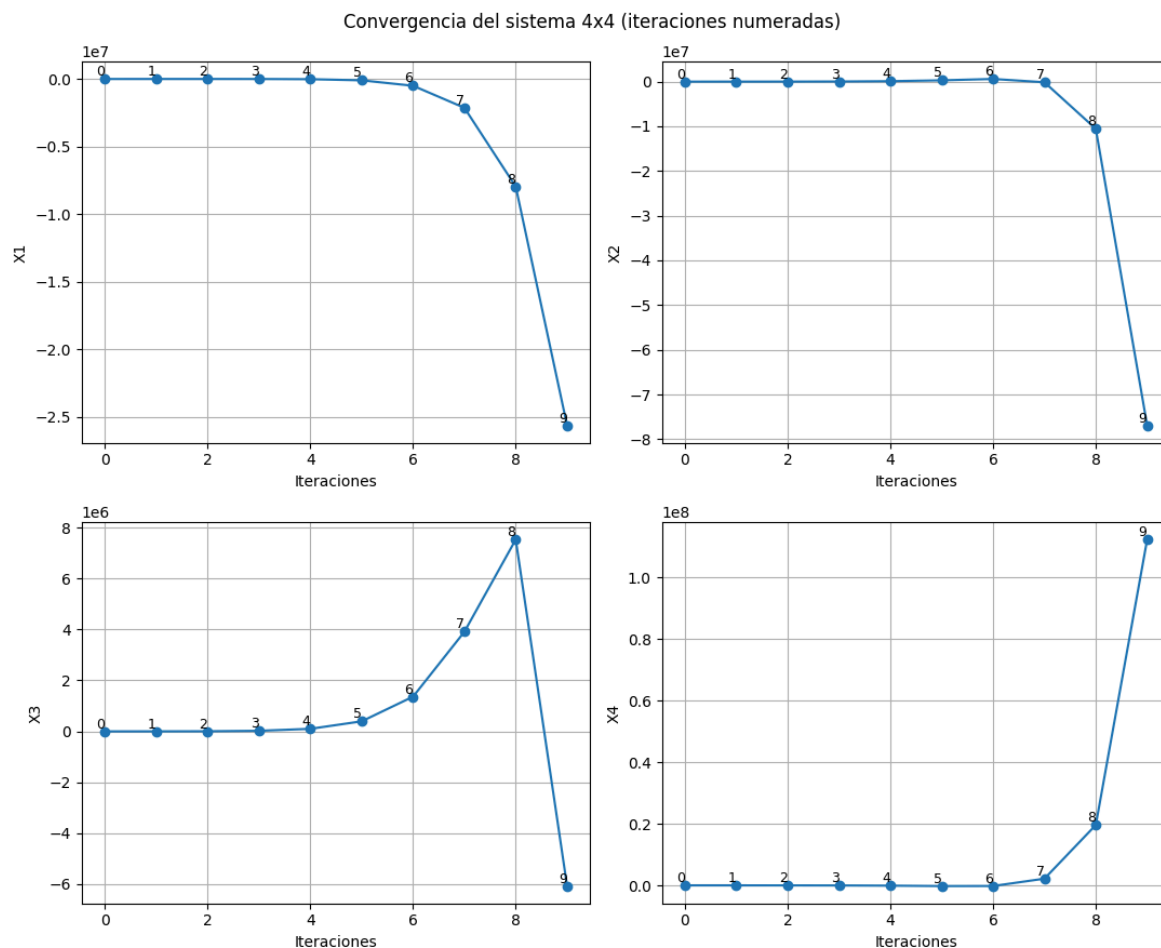
```

```

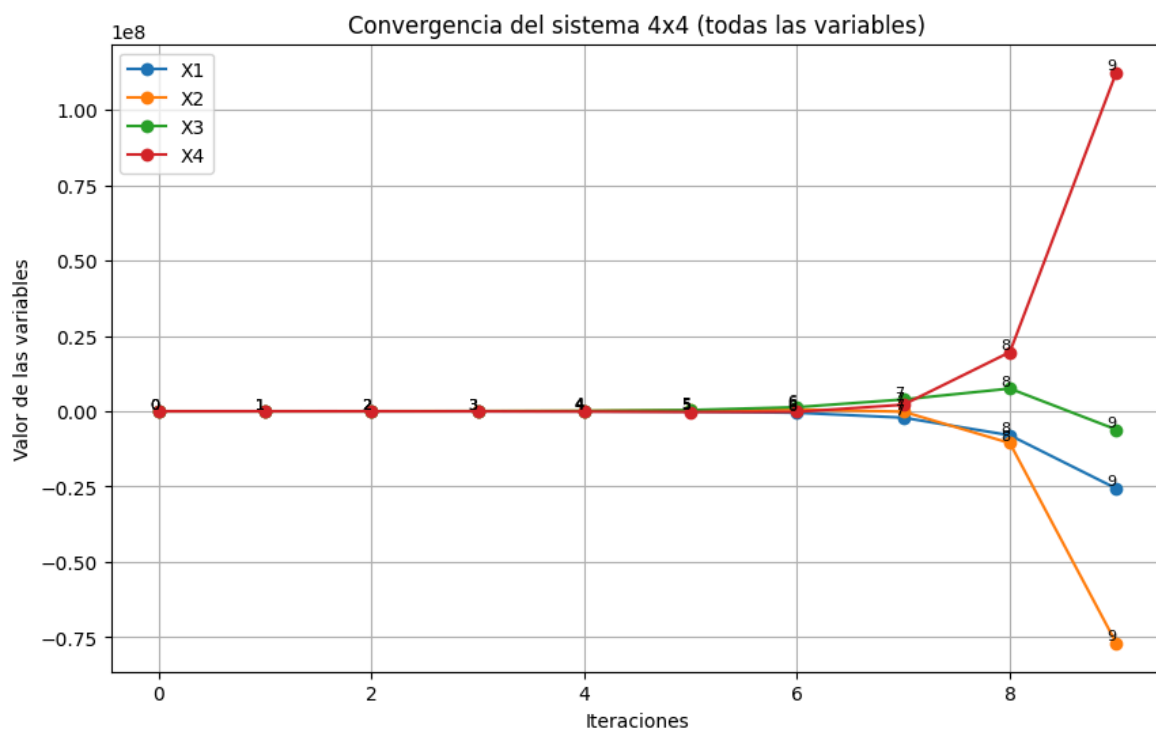
[01-20 21:44:28][INFO][NattyrdGT] i= 0 x: [[2. 1. 0. 3.]]
[01-20 21:44:28][INFO][NattyrdGT] i= 1 x: [[ 1361.   2482.   1113.  -1341.5]]
[01-20 21:44:28][INFO][NattyrdGT] i= 2 x: [[ 1616.   7883.5  4660.  -7670.25]]
[01-20 21:44:28][INFO][NattyrdGT] i= 3 x: [[   -76.5   28724.25  22557.   -299
82.875]]
[01-20 21:44:28][INFO][NattyrdGT] i= 4 x: [[-15029.75   98605.875  100032.5   -
96067.5625]]
[01-20 21:44:28][INFO][NattyrdGT] i= 5 x: [[-100099.125   289266.0625   394746.7
5   -234700.71875]]
[01-20 21:44:28][INFO][NattyrdGT] i= 6 x: [[-498867.4375   601111.46875  136549
5.625   -217094.390625]]
[01-20 21:44:28][INFO][NattyrdGT] i= 7 x: [[-2128519.78125   -110610.984375   39
30684.1875   2133083.5703125]]
[01-20 21:44:28][INFO][NattyrdGT] i= 8 x: [[ -7970619.359375   -10438055.8828125
7637616.40625
19477717.02734375]]
[01-20 21:44:28][INFO][NattyrdGT] i= 1 x: [[ 1361.   2482.   1113.  -1341.5]]
[01-20 21:44:28][INFO][NattyrdGT] i= 2 x: [[ 1616.   7883.5  4660.  -7670.25]]
[01-20 21:44:28][INFO][NattyrdGT] i= 3 x: [[   -76.5   28724.25  22557.   -299
82.875]]
[01-20 21:44:28][INFO][NattyrdGT] i= 4 x: [[-15029.75   98605.875  100032.5   -
96067.5625]]
[01-20 21:44:28][INFO][NattyrdGT] i= 5 x: [[-100099.125   289266.0625   394746.7
5   -234700.71875]]
[01-20 21:44:28][INFO][NattyrdGT] i= 6 x: [[-498867.4375   601111.46875  136549
5.625   -217094.390625]]
[01-20 21:44:28][INFO][NattyrdGT] i= 7 x: [[-2128519.78125   -110610.984375   39
30684.1875   2133083.5703125]]
[01-20 21:44:28][INFO][NattyrdGT] i= 8 x: [[ -7970619.359375   -10438055.8828125
7637616.40625
19477717.02734375]]
[01-20 21:44:28][INFO][NattyrdGT] i= 9 x: [[-2.57119287e+07 -7.65063334e+07 -5.60
340895e+06  1.11959621e+08]]
[01-20 21:44:28][INFO][NattyrdGT] i= 10 x: [[-6.52981555e+07 -4.06779296e+08 -1.6
4222015e+08  5.28059762e+08]]

```

In [78]: `graficar_convergencia_4x4_numerada(historial_x_solutions_seidel21121)`



In [81]: `graficar_4x4_en_un_solo_grafico(historial_x_solutions_seidel21121)`



In [69]: `A = np.array([  
 [ 1, -1, 2, 0],  
 [-1, 1, -1, 3],  
 [ 2, -1, 1, -1],  
 [ 0, 3, -1, 2]  
 ], dtype=float)`



```

b = np.array([
    1360,
    1130,
    1350,
    3650
], dtype=float)
n=1
solution211221, historial_x_solutions_seidel211221 = gauss_seidel(A=A, b=b, x=[0

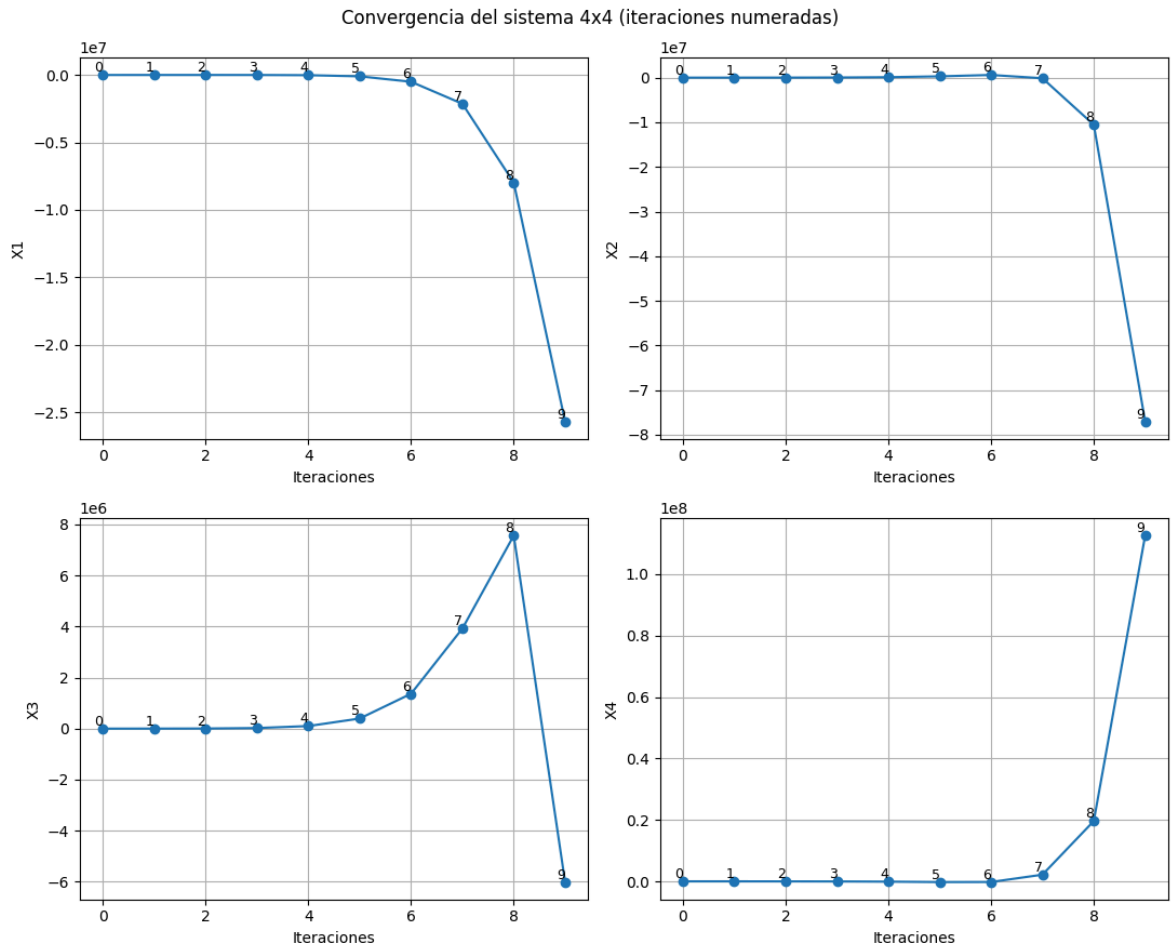
```

```

[01-20 21:51:34][INFO][NattyrdGT] i= 0 x: [[0.25 0.25 0.25 0.25]]
[01-20 21:51:34][INFO][NattyrdGT] i= 1 x: [[ 1359.75   2489.25   1120.    -1348.8
75]]
[01-20 21:51:34][INFO][NattyrdGT] i= 2 x: [[ 1609.25   7905.875   4688.5    -768
9.5625]]
[01-20 21:51:34][INFO][NattyrdGT] i= 3 x: [[  -111.125   28776.0625   22658.75
-30009.71875]]
[01-20 21:51:34][INFO][NattyrdGT] i= 4 x: [[-15181.4375   98636.46875  100339.62
5   -95959.890625]]
[01-20 21:51:34][INFO][NattyrdGT] i= 5 x: [[-100682.78125   288666.515625   3954
22.1875   -233463.6796875]]
[01-20 21:51:34][INFO][NattyrdGT] i= 6 x: [[-500817.859375   596125.3671875   136
5647.40625   -209539.34765625]]
[01-20 21:51:34][INFO][NattyrdGT] i= 4 x: [[-15181.4375   98636.46875  100339.62
5   -95959.890625]]
[01-20 21:51:34][INFO][NattyrdGT] i= 5 x: [[-100682.78125   288666.515625   3954
22.1875   -233463.6796875]]
[01-20 21:51:34][INFO][NattyrdGT] i= 6 x: [[-500817.859375   596125.3671875   136
5647.40625   -209539.34765625]]
[01-20 21:51:34][INFO][NattyrdGT] i= 7 x: [[-2133809.4453125   -138413.99609375
3921015.546875   2169953.76757812]]
[01-20 21:51:34][INFO][NattyrdGT] i= 8 x: [[ -7979085.08984375 -10566800.84570312
7562673.1015625
19633362.81933594]]
[01-20 21:51:34][INFO][NattyrdGT] i= 9 x: [[-2.56907870e+07 -7.70270724e+07 -6.01
078549e+06   1.12537041e+08]]
[01-20 21:51:34][INFO][NattyrdGT] i= 10 x: [[-6.50041414e+07 -4.08624920e+08 -1.6
6078246e+08   5.29900081e+08]]

```

In [74]: `graficar_convergencia_4x4_numerada(historial_x_solutions_seidel211221)`



```
In [75]: import numpy as np
import matplotlib.pyplot as plt

def graficar_4x4_en_un_solo_grafico(historial_x):
    """
    Grafica X1, X2, X3 y X4 en un solo gráfico,
    numerando cada punto según la iteración.

    Parameters
    -----
    historial_x : list
        Lista de vectores x (cada uno de tamaño 4)
    """

    historial_x = np.array(historial_x)

    if historial_x.shape[1] != 4:
        raise ValueError("Este método es solo para sistemas 4x4")

    iteraciones = range(len(historial_x))
    etiquetas = ["X1", "X2", "X3", "X4"]

    plt.figure(figsize=(10, 6))

    for i in range(4):
        plt.plot(iteraciones, historial_x[:, i], marker='o', label=etiquetas[i])

        # Numerar cada punto
        for k in iteraciones:
            plt.text(
                k,
```

```

        historial_x[k, i],
        f'{k}',
        fontsize=8,
        ha='right',
        va='bottom'
    )

plt.xlabel("Iteraciones")
plt.ylabel("Valor de las variables")
plt.title("Convergencia del sistema 4x4 (todas las variables)")
plt.legend()
plt.grid(True)
plt.show()

```

In [77]: graficar\_4x4\_en\_un\_solo\_grafico(historial\_x\_solutions\_seidel211221)

